

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I D.NAVADEEP,192371068 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:D.NAVADEEP

Annexure A

Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n and $\phi(n)$
- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

- a. Encrypt using ECB and CBC modes (Assume IV = 0000)
- b. Compare the ciphertext outputs
- c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison

Given:

- RSA key size = 1024 bits
- ECC key size = 160 bits (equivalent security)
- Device processing capacity = limited (IoT device)
- Analyze computational cost for both systems
- Estimate which consumes less power and memory
- Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

		strong justification			
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. RSA Encryption Analysis

RSA is a public-key cryptography algorithm that uses one public key for encryption and one private key for decryption.

a) Calculate n and $\phi(n)$

Given:

- $p = 11$
- $q = 13$

The modulus is

$$n = 11 \times 13 = 143$$

Euler's Totient Function is

$$\phi(n) = (11-1)(13-1) = 120$$

So,

$$n = 143$$

$$\phi(n) = 120$$

b) Calculate the private key

The public exponent is

$$e = 7$$

We must find d such that

$$7d \equiv 1 \pmod{120}$$

The value satisfying this equation is

$$d = 103$$

Hence,

$$\text{Public Key} = (7, 143)$$

$$\text{Private Key} = (103, 143)$$

c) Encryption

Using

$$C = M^e \pmod{n}$$

Substituting

$$97 \bmod 143 \cdot 97^7 \bmod 143 \cdot 97 \bmod 143$$

The encrypted ciphertext becomes

48

d) Decryption

Using

$$M = C^d \bmod n$$

Substituting

$$48 \cdot 103 \bmod 143 \cdot 48^{103} \bmod 143 \cdot 103 \bmod 143$$

We obtain

9

This confirms that RSA correctly decrypts the original message.

e) Analysis

Changing the value of **e** mainly affects encryption speed.

Smaller values provide faster encryption but must be selected carefully to avoid security risks.

Larger values increase computation time.

Therefore, standard values such as **65537** are widely used because they provide both efficiency and strong security.

2. Block Cipher Mode Evaluation

ECB Encryption

Each block is encrypted independently.

Output:

[0110,0110,0011,0110]

CBC Encryption

Each plaintext block is first combined with the previous ciphertext block.

Output:

[0110,0000,0011,0101]

Comparison

In ECB mode, identical plaintext blocks always produce identical ciphertext blocks.

In CBC mode, repeated plaintext blocks generate different ciphertext blocks because each block depends on the previous encrypted block.

Analysis

CBC offers stronger protection because attackers cannot identify repeated data patterns.

ECB may reveal information about the original plaintext due to repeated ciphertext blocks.

Therefore, CBC is preferred in secure communication systems.

3. DES vs 3-DES vs Blowfish

Encryption Time

Algorithm Time

DES 2000 ms

3-DES 6000 ms

Blowfish 1000 ms

Performance Analysis

DES is fast but outdated because its 56-bit key can be broken using modern computing power.

3-DES improves security by applying DES three times, but this significantly increases encryption time.

Blowfish uses larger keys and performs encryption faster than both DES and 3-DES, making it suitable for modern systems.

Recommendation

For secure real-time communication, Blowfish is the best option because it combines strong security with high-speed encryption and lower computational overhead.

4. Diffie-Hellman Key Exchange

Public Key of User A = **8**

Public Key of User B = **19**

Shared Secret Key = **2**

MITM Attack

A Man-in-the-Middle attacker secretly intercepts the exchanged public keys and replaces them with fake keys. Both users believe they are communicating securely, but in reality, the attacker establishes separate secret keys with each user and gains access to the communication.

Prevention

To prevent this attack, authentication mechanisms such as Digital Signatures, Digital Certificates, and PKI should be used to verify the authenticity of exchanged public keys.

5. ECC vs RSA

RSA requires a much larger key size (1024 bits), which increases computational complexity, memory usage, and power consumption.

ECC provides the same security with only a 160-bit key, making encryption and decryption much faster.

Since IoT devices have limited processing capability, battery life, and memory, ECC is the preferred cryptographic algorithm. It provides strong security while minimizing resource consumption, making it ideal for embedded and mobile systems.